A UNIFORM-EPIGRAPH CASE OF THE UNBOUNDED-DOMAIN QUESTION FOR NEGATIVE SOBOLEV LATTICES

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Status

This packet gives a partial positive answer to the unbounded-domain question in Arora–Glueck–Schwenninger [1]. The source proves its domain theorem for bounded open sets with continuous boundary. Immediately after the bounded-domain construction, the source notes that boundedness was used through compactness of $\overline{\Omega}$ and asks whether a similar result holds for unbounded Ω, perhaps under a uniformity assumption on the continuous boundary. The theorem below proves the same conclusion for a global epigraph whose boundary function is uniformly continuous, retaining the source's generating-cone hypothesis.

Source passage

The source passage, on PDF page 12, reads that the authors do not know whether the bounded-domain result extends to unbounded domains under some uniformity assumption on the continuous boundary.

Then each operator S_n leaves $W^{k,p}(\mathbb{R}^d)$ invariant and the sequence (S_n) converges strongly – both on $L^p(\mathbb{R}^d)$ and on $W^{k,p}(\mathbb{R}^d)$ – to the operator

$$S : f \mapsto f \sum_{\boldsymbol{y} \in Y} h_{\boldsymbol{y}}.$$

From now on we consider $C_c^\infty(\Omega)$ as a subspace of $C_c^\infty(\mathbb{R}^d)$ by extending each function with the value 0 outside of Ω. Similarly, we consider $L^p(\Omega)$ as a subspace of $L^p(\mathbb{R}^d)$. Then S leaves $L^p(\Omega)$ invariant and acts as the identity operator on this space.

Result

Theorem 1. *Let $d \geq 2$, let $k \geq 1$, and let $p, q \in (1, \infty)$ be Holder conjugates. Let $F : \mathbb{R}^{d-1} \to \mathbb{R}$ be uniformly continuous and set*

$$\Omega = \{(x', t) \in \mathbb{R}^{d-1} \times \mathbb{R} : t > F(x')\}.$$

Assume that the positive cone in $W_0^{k,q}(\Omega)$ is generating. Then

$$\operatorname{span}\big(W^{-k,p}(\Omega)_+\big)$$

is a KB-space, and in particular a Banach lattice, with respect to a norm equivalent to the usual span norm. Moreover, the canonical embedding

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$L^p(\Omega) \hookrightarrow W^{-k,p}(\Omega)$ has range a lattice ideal in $\operatorname{span}(W^{-k,p}(\Omega)_+)$ *and is a lattice homomorphism into this span.*

Remark 1. *By applying a rigid motion, and by reflecting the last coordinate, the same statement holds for rigid images of such epigraphs and for corresponding subgraphs. The theorem does not address the separate problem of when $W_0^{k,q}(\Omega)_+$ is generating.*

The approximation operators

We use the source's abstract Theorem 2.1 with $X = W_0^{k,q}(\Omega)$, $Z = L^q(\Omega)$, and $J : X \to Z$ the canonical positive embedding. Thus it is enough to construct positive operators $R_n : Z \to X$ such that $R_n J \to \mathrm{id}_X$ and $JR_n \to \mathrm{id}_Z$ strongly.

Let E denote extension by zero from Ω to $\mathbb{R}^d$. Choose a nonnegative $\rho \in C_c^\infty(\mathbb{R}^d)$ supported in the unit ball, with $\int_{\mathbb{R}^d} \rho = 1$. Choose $\varepsilon_n \downarrow 0$. By uniform continuity of F, choose $\eta_n > 0$ such that

$$|x' - y'| < \eta_n \quad \Longrightarrow \quad |F(x') - F(y')| < \varepsilon_n/4.$$

Set

$$r_n = \min(\eta_n, \varepsilon_n/4), \qquad \rho_n(x) = r_n^{-d}\rho(x/r_n).$$

For $f \in L^q(\Omega)$ define, on $\mathbb{R}^d$,

$$T_n f = \rho_n * \big(Ef(\,\cdot - \varepsilon_n e_d)\big),$$

and let $R_n f$ be the restriction of $T_n f$ to Ω.

Lemma 1. *For every n and every $f \in L^q(\Omega)$, the function $T_n f$ vanishes outside Ω. Moreover $R_n f \in W_0^{k,q}(\Omega)$ and $R_n : L^q(\Omega) \to W_0^{k,q}(\Omega)$ is positive and bounded.*

Proof. Suppose that $T_n f(x) \neq 0$. Then for some $z \in \Omega$ one has $|x - (z + \varepsilon_n e_d)| < r_n$. Hence $|x' - z'| < r_n \le \eta_n$ and

$$x_d > z_d + \varepsilon_n - r_n > F(z') + \varepsilon_n - r_n \ge F(x') + \varepsilon_n - r_n - \varepsilon_n/4 \ge F(x') + \varepsilon_n/2.$$

Thus $x \in \Omega$. So the zero extension of $R_n f$ is exactly $T_n f$.

Convolution with ρ_n maps $L^q(\mathbb{R}^d)$ boundedly into $W^{k,q}(\mathbb{R}^d)$, with a constant depending on n. Therefore $T_n f \in W^{k,q}(\mathbb{R}^d)$, and the preceding paragraph shows that its restriction is supported inside Ω. For each compact ball cutoff $\chi_m \in C_c^\infty(\mathbb{R}^d)$ with $\chi_m \to 1$, the function $\chi_m T_n f$ is in $C_c^\infty(\Omega)$: its compact support lies in the closed set $x_d \ge F(x') + \varepsilon_n/2$, hence in Ω. The usual cutoff convergence in $W^{k,q}(\mathbb{R}^d)$ gives $\chi_m T_n f \to T_n f$. Hence $R_n f \in W_0^{k,q}(\Omega)$. Positivity is immediate from $\rho_n \ge 0$ and from positive zero extension and translation. $\square$

Lemma 2. *The operators above satisfy*

$$JR_n \to \mathrm{id}_{L^q(\Omega)} \quad \textit{strongly on } L^q(\Omega)$$

and

$$R_n J \to \mathrm{id}_{W_0^{k,q}(\Omega)} \quad \textit{strongly on } W_0^{k,q}(\Omega).$$

Proof. For $f \in L^q(\Omega)$, the zero extension Ef belongs to $L^q(\mathbb{R}^d)$. Since $\varepsilon_n \to 0$ and $r_n \to 0$, translation followed by the standard approximate identity ρ_n converges to the identity in $L^q(\mathbb{R}^d)$. The restriction to Ω gives $JR_n f \to f$ in $L^q(\Omega)$.

If $u \in W_0^{k,q}(\Omega)$, then its zero extension Eu belongs to $W^{k,q}(\mathbb{R}^d)$ by the definition of $W_0^{k,q}(\Omega)$. Translation and mollification converge to the identity in $W^{k,q}(\mathbb{R}^d)$. Since the zero extension of $R_n Ju$ is $T_n u$, Lemma 1 yields $R_n Ju \to u$ in $W_0^{k,q}(\Omega)$. □

Proof of Theorem 1

The space $X = W_0^{k,q}(\Omega)$ is reflexive, since $q \in (1, \infty)$ and it is a closed subspace of the reflexive Sobolev space defined by zero extension. By hypothesis, X_+ is generating. The space $Z = L^q(\Omega)$ is a Banach lattice, and $J : X \to Z$ is positive. Lemmas 1 and 2 provide positive operators $R_n : Z \to X$ for which both $R_n J \to \mathrm{id}_X$ and $JR_n \to \mathrm{id}_Z$ strongly, hence also weakly.

The source's Theorem 2.1 therefore applies. It says that $\operatorname{span}(X'_+)$ is a monotonically complete Banach lattice under an equivalent span norm, that it is a KB-space because X is reflexive, and that $J'Z'$ is a lattice ideal in $\operatorname{span}(X'_+)$. Identifying $X' = W^{-k,p}(\Omega)$ and $Z' = L^p(\Omega)$ gives precisely the conclusion of the theorem.

Scope and verification notes

The proof only replaces the bounded-domain compactness step in the source. It does not prove that $W_0^{k,q}(\Omega)_+$ is generating, and so it leaves the source's later generating-cone open problem untouched. It also uses the global uniformly continuous epigraph structure in an essential way: the single upward translation is what provides a uniform boundary buffer for all spatial locations.

Cheap local indexes were searched for the arXiv id and for the terms "negative Sobolev", "unbounded", "uniformly continuous boundary", and "epigraph". The only local content hit for this question was the source passage itself; other hits were bibliographic citations or unrelated uses of epigraph terminology.

References

[1] S. Arora, J. Glueck, and F. L. Schwenninger, *The lattice structure of negative Sobolev and extrapolation spaces*, arXiv:2404.02116.